Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9999363422242026	0.8434728164457894 – 1.0342578520770012

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.3391463976714771	0.3347444354363207 – 0.34212995784873657

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

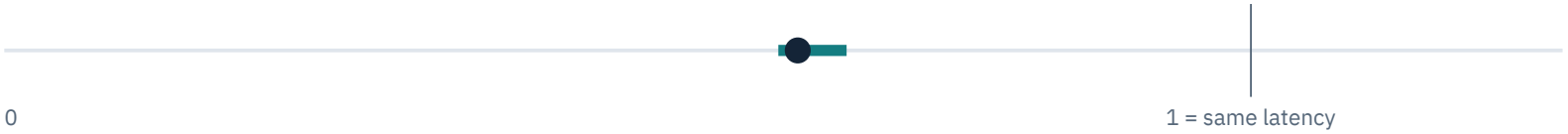
Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.3395659435871721	0.29953666777081356 – 0.3520476746781338

Thin QR

512 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.6364933793072876	0.6209107889725535 – 0.6756612892776158

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.5251560195707211	0.4995823827409169 – 0.533394334068534

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0065869156092198	0.9334875283446712 – 1.0366644664466447

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9692970033422081 – 1.0242323200992556

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	1.0203252032520325	95% INTERVAL	0.9828453796889296 – 1.0506988647316935

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE **1**

95% INTERVAL 0.9642077695329551 – 1.0186666666666666

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE **1.0341187136270142**

95% INTERVAL 0.9979659367396594 – 1.0490804116484436

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9852494088503547	0.9299589318823753 – 1.0053293424102498

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9968545275313336	0.9725039674746369 – 1.0434932624259767

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9854261083743843	0.9696438041145655 – 0.995589617708063

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.98656	0.9482543640897756 – 1.1184085885696242

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9959645031113004	0.9219876288659794 – 1.0164903225806452

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0047169811320755	0.9952867538931413 – 1.039894165994258

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.8290898448931812 – 1.0305454545454547

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	1.0229132569558101	95% INTERVAL	0.9333333333333333 – 1.0463169642857142

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.995259977566306	0.9373491217847332 – 1.0350073578595318

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9960448117608732	0.9707239802826404 – 1.0095877952453927

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9951801061606067	0.9423570914820015 – 1.0506741840165743

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0216484838542095	0.5988323603002502 – 1.2320360685302074

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0137858548492327	0.7405825022956959 – 1.0541735305919873

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0029569620253165	0.9843949367088608 – 2.0068221741084193